

THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

AWARD TEAM HOMEWORK SET #7

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2017
Professor: Dr. Michael Caracotsios
E-mail: mcaracot@uic.edu

Your name: Vrushant Patel
Your name: Harshik Patel
Your name: _____

Due Date: Wednesday Nov-22, 2017 in class

Number of points100
Your score

Award: Gift card to each member of the winning team

To be considered for the award, the team homework final report must be typed. Homework results should be presented in a logical and concise manner, with all steps in derivation processes [if any] shown very clearly. The VBA code must be well documented and attached to the homework report. Images of the pertinent Excel worksheets must be clearly shown in the homework solution report. You may also want to include anything else (like external references) you believe will increase your chances of winning the award.

Remember: All members of the team that wins a total of three(3) awards will be given 100 points towards the final examination and will be excused from taking it.

TABLE OF CONTENTS

Serial No.	Topic	Page no.
1	Description of the project posed	3 - 3
2	Procedure to solve for density using methodology I	4 - 7
3	Results from MS Excel	8 - 8
4	Parity plot and Re-estimation of Rackett factor for mixture	9 - 10
5	Procedure to solve for density using methodology II	10 - 12
6	Results for temperature = 25 degree Celsius	13 - 13
7	Results for temperature = 30 degree Celsius	14 - 14
8	Results for temperature = 35 degree Celsius	15 - 15
9	Results for temperature = 40 degree Celsius	16 - 16
10	Proposal of parameters at 50 degree Celsius using methodology II	17 - 17
11	Thermodynamic Consistency	18 - 19
12	Final proposal to XYZ corporation	20 - 20

Homework #7: Density of Methanol-water liquid mixtures

Scenario

It is a project given by the XYZ corporation to develop a methodology for the calculation of the density of methanol-liquid mixtures at 50 degree Celsius. Below is some experimental data for the different temperatures and mixtures compositions

	T=25.0°C	T=30.0°C	T=35.0°C	T=40.0°C
Mole % Methanol	Volume ml/mol	Volume ml/mol	Volume ml/mol	Volume ml/mol
0.000	18.068	18.094	18.123	18.156
5.880	19.221	19.253	19.288	19.330
12.330	20.468	20.508	20.555	20.607
19.430	21.852	21.909	21.972	22.042
27.270	23.447	23.523	23.602	23.689
35.990	25.293	25.357	25.457	25.565
45.760	27.442	27.548	27.657	27.783
56.750	29.951	30.076	30.223	30.371
69.220	32.905	33.081	33.272	33.440
83.500	36.468	36.657	36.898	37.096
91.440	38.524	38.737	38.987	39.220
100.000	40.720	40.981	41.234	41.490

The two methodologies to investigate are:

- 1) Apply the Rackett Equation for the density prediction. Consider re-estimating the Rackett ZA factor for methanol-water mixture.
- 2) Develop a model for the Excess Volume V^E using the thermodynamic principles and estimate the parameters of the model from the experimental data using multiple linear regression.

Methodology I:

Following were the formulas provided to solve Rackett mixture liquid volume equation

$$V^l = \frac{RT_{cm}}{P_{cm}} (Z_m^{RA}) \left[1 + (1 - T_r)^{\frac{2}{7}} \right]$$

$$T_{cm} = \sum_i \sum_j \frac{x_i x_j V_{ci} V_{cj} (T_{ci} T_{cj})^{\frac{1}{2}} (1 - k_{ij})}{V_{cm}^2}$$

$$\frac{T_{cm}}{P_{cm}} = \sum_i x_i \frac{T_{ci}}{P_{ci}}$$

$$Z_m^{RA} = \sum_i x_i Z_i^{RA}$$

$$V_{cm} = \sum_i x_i V_{ci}$$

$$V_{cm} = \sum_i x_i V_{ci}$$

$$T_r = \frac{T}{T_{cm}}$$

$$k_{ij} = 1 - \frac{8(V_{ci} V_{cj})^{1/2}}{\left(V_{ci}^{\frac{1}{3}} + V_{cj}^{\frac{1}{3}} \right)^3}$$

NOTE: Everything in the formula is in SI units. P [=] Pascal's, T [=] Kelvin and V [=] cm³/mol

T_{cm}	Critical Temperature of the mixture
V_{cm}	Critical Volume of the mixture
P_{cm}	Critical Pressure of the mixture
Z_m^{RA}	Rackett factor for the mixture
k_{ij}	Interaction parameter
x_i	Mole fraction of the pure component i
T_{ci}	Critical Temperature of the pure component i
V_{ci}	Critical Volume of the pure component i
P_{ci}	Critical Pressure of the pure component i
Z_i^{RA}	Rackett factor of the pure component i

Procedure:

Consider using the following indices 1 for methanol and 2 for water

Step 1: Transfer the physical properties for the methanol and water from the Aspen Plus Software, and convert the pressure and temperature in the units of Pa and K respectively.

Step 2: Consider expanding the formula to check what needs to be calculated to solve the problem

- 1) For T_{cm} equation there would be total of 4 terms, there are two components (namely methanol and water) so the indices i and j will range from 1 to 2 which will result in following equation

- For i = 1 and j = 1, the critical temperatures will get squared and since it will be under square root only one of them will remain. While, the critical volume will get squared leading to following term

$$\frac{x_1^2 * V_{c1}^2 * T_{c1}(1 - k_{11})}{V_{cm}^2}$$

- For i = 1 and j = 2 and for i = 2 and j = 1 the expression would remain the same, since everything is just multiplied, and it does not matter in which order it is being multiplied.

$$\frac{x_1 * x_2 * V_{c1} * V_{c2} * (T_{c1} * T_{c2})^{\frac{1}{2}}(1 - k_{12})}{V_{cm}^2}$$

- For i = 2 and j = 2, the critical temperatures will get squared and since it will be under square root only one of them will remain. While, the critical volume and mole fraction will get squared leading to following term

$$\frac{x_2^2 * V_{c2}^2 * T_{c2}(1 - k_{22})}{V_{cm}^2}$$

Thus, the final T_{cm} expression will be (NOTE: $k_{11} = k_{22} = 0$ which is shown in next step).

$$T_{cm} = \frac{x_1^2 * V_{c1}^2 * T_{c1}}{V_{cm}^2} + \frac{x_2^2 * V_{c2}^2 * T_{c2}}{V_{cm}^2} + 2 \frac{x_1 * x_2 * V_{c1} * V_{c2} * (T_{c1} * T_{c2})^{\frac{1}{2}}(1 - k_{12})}{V_{cm}^2}$$

2) Consider expanding the interaction parameter k_{ij}

- For $i = 1$ and $j = 1$. Similar formula would develop for $i = 2$ and $j = 2$.

$$k_{11} = 1 - \frac{8(V_{c1}V_{c1})^{\frac{1}{2}}}{\left(V_{c1}^{\frac{1}{3}} + V_{c1}^{\frac{1}{3}}\right)^3}$$

We simplify the expression to result in following

$$k_{11} = 1 - \frac{8(V_{c1}^2)^{\frac{1}{2}}}{\left(2V_{c1}^{\frac{1}{3}}\right)^3}$$

From the above expression, in the numerator square root of a squared term will result in only one critical volume. While, in the denominator a factor of 8 will appear multiplied by critical volume (since factor of 3 will cancel out) to get following expression

$$k_{11} = 1 - \frac{8V_{c1}}{8V_{c1}} = 0$$

NOTE: This is what is expected as there is no interaction between similar compounds.

$$\text{Hence, } k_{11} = k_{22} = 0.$$

- For $i = 1$ and $j = 2$. Similar formula would develop for $i = 2$ and $j = 1$

$$k_{12} = 1 - \frac{8(V_{c1}V_{c2})^{\frac{1}{2}}}{\left(V_{c1}^{\frac{1}{3}} + V_{c2}^{\frac{1}{3}}\right)^3}$$

3) Consider expanding the V_{cm} expression

$$V_{cm} = x_1V_{c1} + x_2V_{c2}$$

4) Consider expanding the Z_m^{RA} expression

$$Z_m^{RA} = x_1Z_1^{RA} + x_2Z_2^{RA}$$

5) Finally, expanding the T_{cm} / P_{cm} equation

$$\frac{T_{cm}}{P_{cm}} = x_1 \frac{T_{c1}}{P_{c1}} + x_2 \frac{T_{c2}}{P_{c2}}$$

Step 3: Using the formula developed in Step 2 and the given information, Excel spreadsheet was created to calculate the liquid volume of the mixture using the Rackett equation.

The value of universal gas constant R used was: $8.314 \times 10^6 \text{ Pa cm}^3 \text{ K}^{-1} \text{ mol}^{-1}$

Step 4: Density of the mixture was calculated by taking the inverse of the liquid volume mixture and converting into units of g/cm^3 by multiply the density in mol/cm^3 by the molecular weight of the mixture which is calculated using the formula shown below.

$$\text{MW of the mixture} = x_1 \text{MW}_1 + x_2 \text{MW}_2$$

Step 5: The actual density of the pure components was calculated from the given volume of the pure components.

At 25 degree Celsius:

Density of pure methanol (For $x_1 = 1$)

$$\frac{1 \text{ mol}}{40.72 \text{ cm}^3} * 32.0422 \frac{\text{g}}{\text{mol}} = 0.7869 \frac{\text{g}}{\text{cm}^3}$$

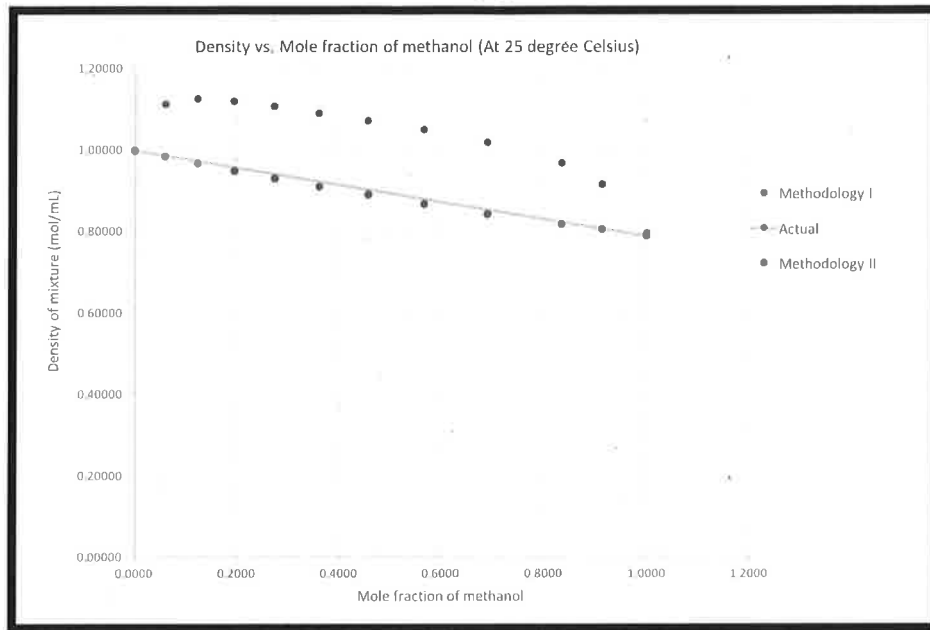
Density of pure water (For $x_1 = 0$)

$$\frac{1 \text{ mol}}{18.068 \text{ cm}^3} * 18.0153 \frac{\text{g}}{\text{mol}} = 0.9971 \frac{\text{g}}{\text{cm}^3}$$

Step 6: Lastly, a parity plot was created for the density of mixture (g/cm^3) vs. Mole fraction of methanol using the values of density of pure methanol and water calculated in the above step as the reference line to check for the deviation.

It is found that solving for volume using the Rackett equation is not accurate to predict the densities as seen from the deviation of the data points from the actual.

Parity plot



- In the graph, it is evident that Methodology I (Rackett factor calculations) is not accurate enough to predict the densities, whereas the Methodology II (Using multi-linear regression) proves to be rather more accurate method to predict the densities of methanol-water mixture at various molar compositions, as its data points are scattered very closely to the actual values.

Re-estimating the value of Z_{RA}

- To re-estimate the value of Z_{RA} of methanol-water mixture, we used the density values from Methodology II and created a column of the residual squared using the density calculated from Method I and Method II to calculate the sum of the residual square. (we used the values at 25 degree Celsius, although any temperature is fine).
- Solver was then used to minimize the sum of the residual squared error by changing the array of $Z_m(RA)$ values.
- It turns out that the values of $Z_m(RA)$ are slightly higher than that calculated values using the formula provided for the $Z_m(RA)$ in case of Rackett equation.

The results are shown below are shown on next page.

Zm(RA)	Zm(RA) -estimated
0.2432	0.2428
0.2428	0.2589
0.2423	0.2620
0.2418	0.2631
0.2413	0.2636
0.2406	0.2638
0.2400	0.2639
0.2392	0.2638
0.2383	0.2629
0.2373	0.2594
0.2368	0.2540
0.2362	0.2372

Methodology 2:

Developing a model for Excess Volume V^E using the thermodynamic principles to estimate the parameters of the model at 50 degree Celsius from the experimental data using multi-linear regression.

NOTE: Once following steps are implemented for 25 degree Celsius, same procedure applies for other temperature.

Step 1: Using the data for the volume of pure methanol (V_1) and pure water (V_2) ideal volume of the mixture for all molar compositions is being calculated from the formula below

$$\text{Ideal Volume of the mixture } V^{ID} = x_1V_1 + x_2V_2$$

Step 2: Using the given volume data and the calculated ideal volume of the mixture, excess volume is calculated for all molar compositions from the formula below

$$\text{(Actual) Excess Volume of the mixture } V^E = V - V^{IG}$$

Step 3: A column for the estimated excess volume is created where in excess volume will be estimated from the parameters using first the simplest formula for excess volume.

$$\text{(Estimated) Excess Volume of the mixture } V^E = x_1 x_2 A_0$$

If the one parameter is not enough to estimate the model, then additional parameters are added in the form

$$\text{(Estimated) Excess Volume of the mixture } V^E = x_1 x_2 [A_0 + A_1 x_1 + A_2 x_1^2 + \dots]$$

But, since parabola is very much symmetrical one parameter is adequate to predict an equation for excess volume.

Step 4: A graph is created between Excess volume (cm^3/mol) vs. Mole fraction of the methanol on which two data series are being plotted i.e. (Actual) Excess Volume and (Estimated) Excess Volume to monitor whether the estimated parameter proves to be a good guess or not.

Step 5: In addition, another column for the squared residual is being added whose total sum is being calculated to estimate the parameter that appropriately minimizes the error.

Step 6: Once manual estimation of parameters was done to predict value which would minimize the error, Solver was automated to get accurate results by minimizing the sum of the residual squared error by changing A_0 parameter of regression.

Solver Parameters

Set Objective:

To: ☐ Max ☒ Min ☐ Value Of:

By Changing Variable Cells:

Subject to the Constraints:

☐ Make Unconstrained Variables Non-Negative

Select a Solving Method:

Solving Method
Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

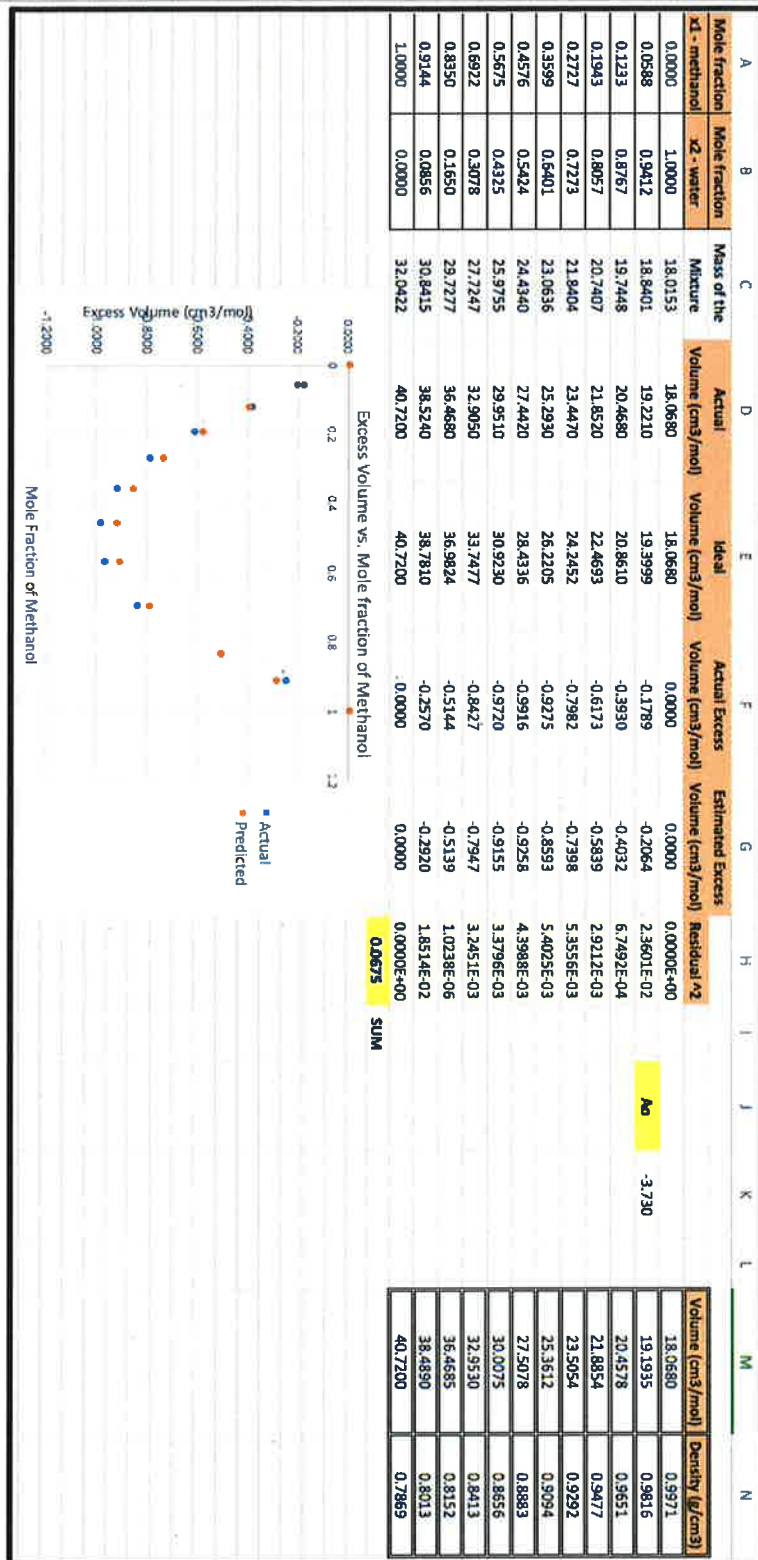
Step 7: After the parameter is solved using Solver, volume is being calculated by adding the Ideal volume calculated earlier to the (Estimated excess) volume which is predicted using the parameter solved using solver.

Step 8: Density is calculated by taking the inverse of the volume and multiplied by the molecular weight of the mixture to obtain density in units of g/cm^3 .

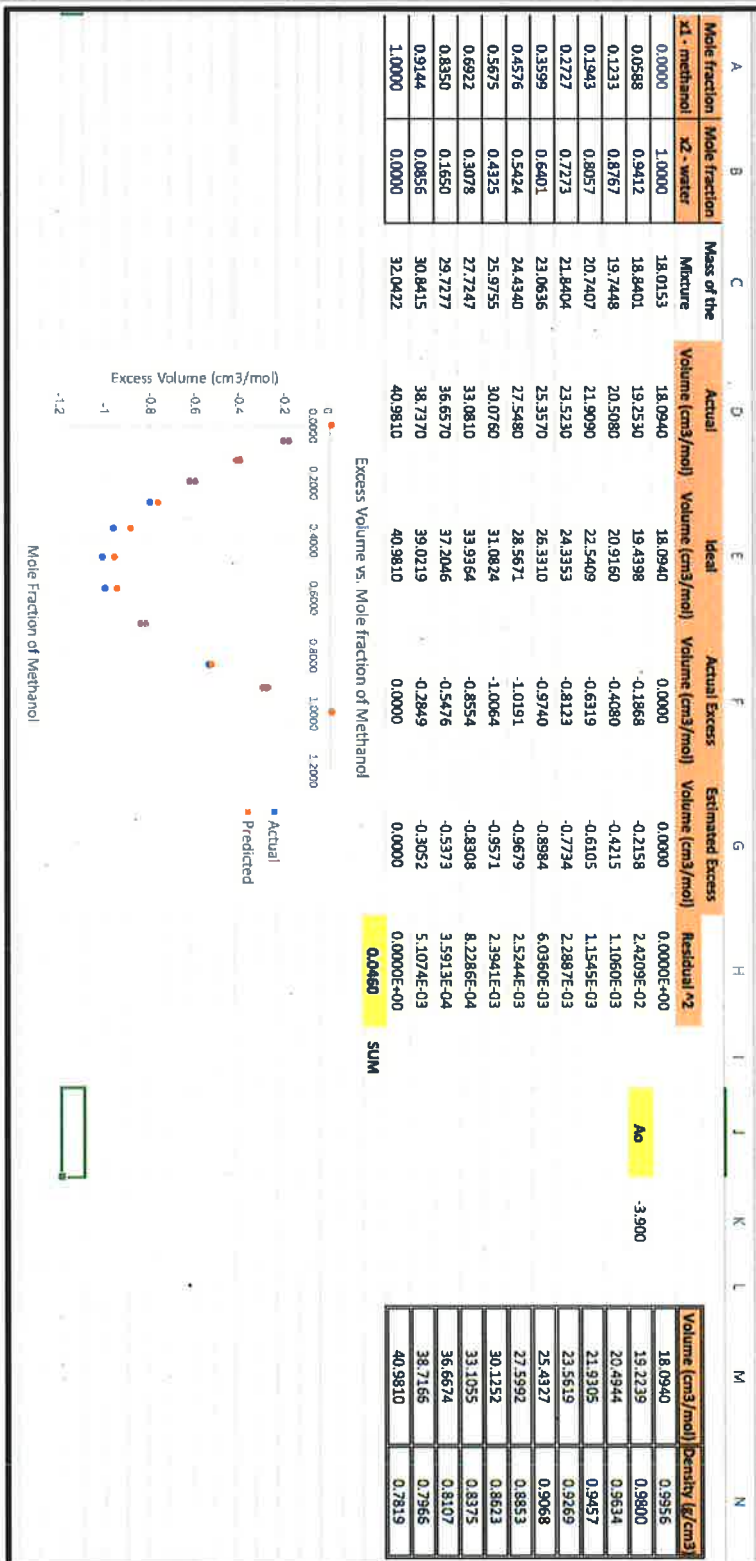
Step 9: Lastly, a relationship is being developed for the parameter A_o and the temperature by plotting a graph A_o vs. temperature to predict a value for A_o at 50 degree Celsius.

The results for values at different temperature are shown below.

For 25 degree, Celsius



For 30 degree Celsius



For 35 degree Celsius

A	B	C	D	E	F	G	H	I	J	K	L	M	N
Mole fraction x1 - methanol	Mole fraction x2 - water	Mass of the Mixture	Actual Volume (cm ³ /mol)	Ideal Volume (cm ³ /mol)	Actual Excess Volume (cm ³ /mol)	Estimated Excess Volume (cm ³ /mol)	Residual *2						
0.0000	1.0000	18.0153	18.1230	18.1230	0.0000	0.0000	0.0000E+00						
0.0588	0.9412	18.8401	19.2880	19.4819	-0.1939	-0.2159	1.2842E-02			-3.901			
0.1233	0.8767	19.7448	20.5550	20.9726	-0.4176	-0.4217	9.7485E-05						
0.1943	0.8057	20.7407	21.9720	22.6135	-0.6415	-0.6107	2.2969E-03						
0.2727	0.7273	21.8404	23.6020	24.4254	-0.8234	-0.7737	3.6325E-03						
0.3599	0.6401	23.0636	25.4570	26.4405	-0.9836	-0.8987	7.4531E-03						
0.4576	0.5424	24.4340	27.6570	28.6986	-1.0416	-0.9683	4.9530E-03						
0.5875	0.4125	25.9755	30.2230	31.2385	-1.0155	-0.9575	3.2582E-03						
0.6922	0.3078	27.7247	33.2720	34.1204	-0.8484	-0.8312	4.1319E-04						
0.8350	0.1650	29.7277	36.9890	37.4207	-0.5227	-0.5375	8.0218E-04						
0.9144	0.0856	30.8415	38.9870	39.2557	-0.2687	-0.3054	1.8614E-02						
1.0000	0.0000	32.0422	41.2340	41.2340	0.0000	0.0000	0.0000E+00						
Excess Volume vs. Mole fraction of Methanol							0.0544	SUM					

Excess Volume (cm³/mol)

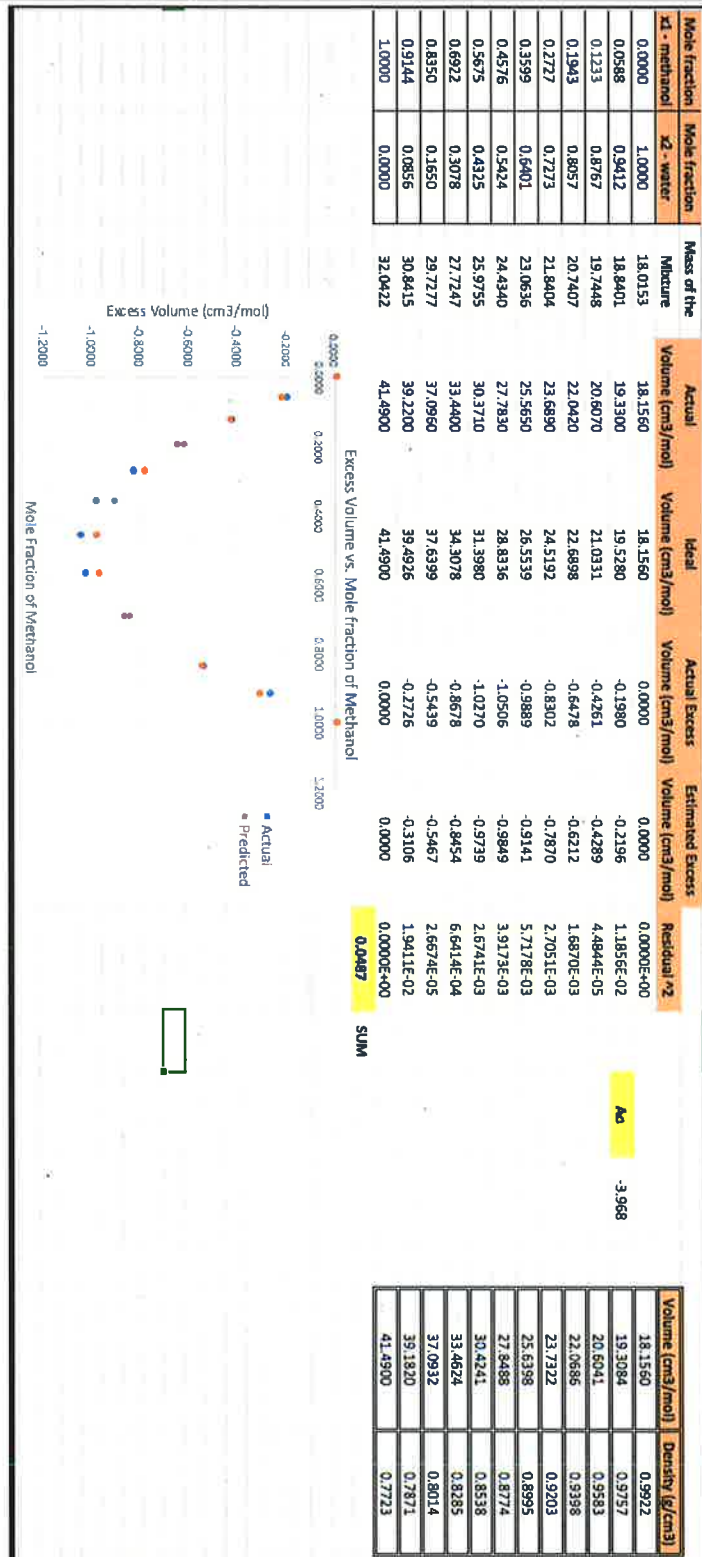
Mole Fraction of Methanol

Actual

Predicted

Volume (cm ³ /mol)	Density (g/cm ³)
18.1230	0.9941
19.2880	0.9779
20.5550	0.9608
21.9720	0.9426
23.6020	0.9234
25.4519	0.9030
27.7303	0.8811
30.2810	0.8578
33.2892	0.8328
36.8832	0.8060
38.9503	0.7918
41.2340	0.7711

For 40 degree Celsius

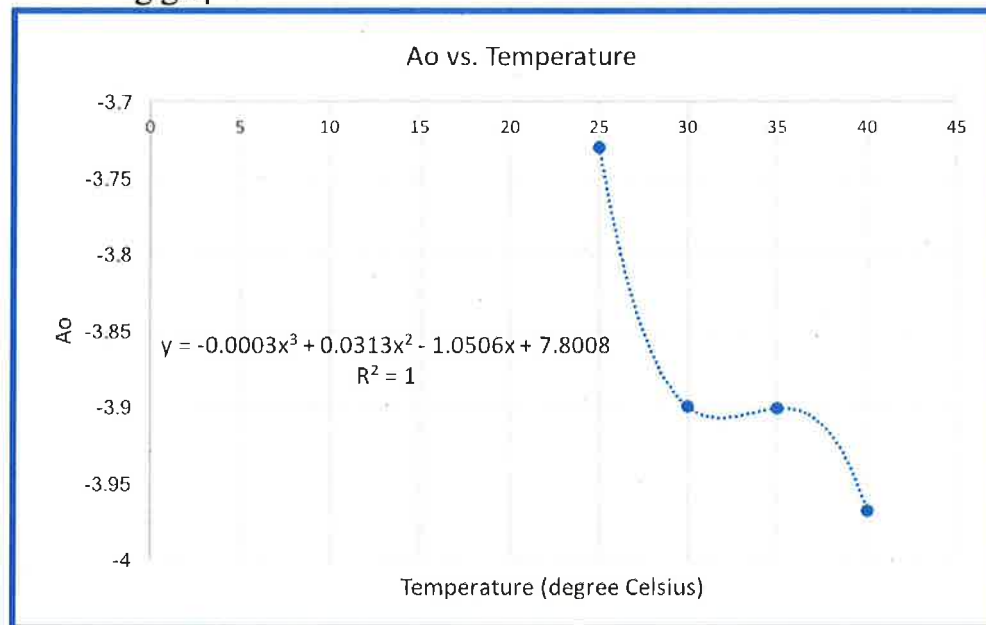


Proposal of parameter at 50 degree Celsius

The following is the regression parameter data at different temperatures from the experimental

REGRESSION PARAMETERS					
TEMPERATURE	25	30	35	40	50
Ao	-3.730	-3.900	-3.901	-3.968	-3.979

- To predict the parameter Ao at 50 degree Celsius we used the means of graphical method to establish the relationship between Ao and Temperature.
- Thus, a plot was created between Ao vs. Temperature with temperature being the independent variable. Hence
- Thereafter, we fitted the data points using a third-order polynomial which resulted in following graph



Using the trend line equation, we solved for Ao by plugging in 50 for the temperature, thus resulting in Ao to be -3.979 at 50 degree Celsius.

Therefore, the final model for excess volume at 50 degree Celsius being predicted is

$$\text{(Estimated) Excess Volume of the mixture } V^E = -3.979x_1x_2$$

Showing Thermodynamic Consistency

The equation that needs to be proven is: $x_1 d\bar{V}_1 + x_2 d\bar{V}_2 = 0$

Where

$$\bar{V}_1 = V - x_2 \frac{dV}{dx_2}$$

$$\bar{V}_2 = V - x_1 \frac{dV}{dx_1}$$

$$V = V^{ID} + V^E$$

$$V = x_1 V_1 + x_2 V_2 + V^E$$

where V_1 is the volume of the pure component (methanol) and V_2 is the volume of the pure component (water).

We predicted the mode for $V^E = A_0 x_1 x_2$

Hence, we can write the total volume of the mixture as: $V = x_1 V_1 + x_2 V_2 + A_0 x_1 x_2$

- Eliminate x_1 by replacing it with $1 - x_2$

$$V = (1 - x_2)V_1 + x_2 V_2 + A_0(1 - x_2)x_2$$

$$\frac{dV}{dx_2} = -V_1 + V_2 + A_0 - 2A_0 x_2 = -V_1 + V_2 + A(1 - 2x_2)$$

$$\bar{V}_1 = x_1 V_1 + x_2 V_2 + A_0 x_1 x_2 - x_2 [-V_1 + V_2 + A(1 - 2x_2)]$$

$$\bar{V}_1 = x_1 V_1 + x_2 V_2 + A_0 x_1 x_2 + V_1 x_2 - V_2 x_2 - A x_2 (1 - 2x_2)$$

$$\bar{V}_1 = V_1(x_1 + x_2) + A_0(x_1 x_2 - x_2 + 2x_2^2)$$

$$d\bar{V}_1 = A_0(x_1 dx_2 + x_2 dx_1 - dx_2 + 4x_2 dx_2)$$

- Eliminate x_2 by replacing it with $1 - x_1$

$$V = x_1V_1 + (1 - x_1)V_2 + A_0(1 - x_1)x_1$$

$$\frac{dV}{dx_1} = -V_2 + V_1 + A_0 - 2A_0x_1 = -V_2 + V_1 + A(1 - 2x_1)$$

$$\overline{V}_2 = x_1V_1 + x_2V_2 + A_0x_1x_2 - x_1[-V_2 + V_1 + A(1 - 2x_1)]$$

$$\overline{V}_2 = x_1V_1 + x_2V_2 + A_0x_1x_2 + V_2x_1 - V_1x_1 - Ax_1(1 - 2x_1)]$$

$$\overline{V}_2 = V_2(x_1 + x_2) + A_0(x_1x_2 - x_1 + 2x_1^2)$$

$$d\overline{V}_2 = A_0(x_1dx_2 + x_2dx_1 - dx_1 + 4x_1dx_1)$$

Solving for LHS of the thermodynamic consistency equation

$$x_1d\overline{V}_1 + x_2d\overline{V}_2 = x_1(A_0(x_1dx_2 + x_2dx_1 - dx_2 + 4x_2dx_2)) + x_2(A_0(x_1dx_2 + x_2dx_1 - dx_1 + 4x_1dx_1))$$

$$x_1(A_0x_1dx_2 + A_0x_2dx_1 - A_0dx_2 + 4A_0x_2dx_2) + x_2(A_0x_1dx_2 + A_0x_2dx_1 - A_0dx_1 + 4A_0x_1dx_1)$$

$$(A_0x_1^2dx_2 + A_0x_1x_2dx_1 - A_0x_1dx_2 + 4A_0x_1x_2dx_2) + (A_0x_1x_2dx_2 + A_0x_2^2dx_1 - A_0x_2dx_1 + 4A_0x_1x_2dx_2)$$

Consider replacing $dx_1 = -dx_2$

$$(A_0x_1^2dx_2 - A_0x_1x_2dx_2 - A_0x_1dx_2 + 4A_0x_1x_2dx_2) + (A_0x_1x_2dx_2 - A_0x_2^2dx_2 + A_0x_2dx_2 - 4A_0x_1x_2dx_2)$$

The red terms will cancel out, thus the resultant equation would be

$$A_0dx_2(x_1^2 - x_2^2 + x_2 - x_1)$$

Consider replacing $x_2 = 1 - x_1$

$$A_0dx_2(x_1^2 - (1 - x_1)^2 + 1 - x_1 - x_1)$$

$$A_0dx_2(x_1^2 - (1 + x_1^2 - 2x_1) + 1 - x_1 - x_1)$$

$$A_0dx_2(x_1^2 - 1 - x_1^2 + 2x_1 + 1 - 2x_1)$$

Everything inside the brackets will cancel out, thus leading to our RHS = 0

Q.E.D

Final Proposal to XYZ corporation

From the analysis of two methodologies, the first being using Rackett formula for mixture Liquid volume to predict the densities of methanol-water mixture and second being using multi-linear regression to find unknown fitting parameter to predict an equation for excess volume at 50 degree Celsius.

We would recommend using the later to predict the densities of methanol-water mixture, as the results are very closer to the actual values and is prone to less error rather than using Rackett formula which gives densities that are very far deviated from the actual values.

For the first method, the results are deviated because the given formula for the Rackett factor for the mixture does not provide a good estimation of the density, because of which densities of methanol-water mixture calculated are not accurate.

However, if we wish to use Method I to calculate densities then using we need to re-estimate Rackett factors.

References

- Lecture Block #6 – Professor Michael Caracotsios Notes
- Lecture Block #7 – Professor Michael Caracotsios Notes